

AAKASH S S

Chennai, Tamil Nadu

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Executive Summary

AI & Data Science undergraduate with hands-on experience across the full ML lifecycle, from data preprocessing and model development to deployment, gained through academic projects and industry internships. Experienced in building practical AI and data-driven solutions that address real-world problems, with measurable results. A fast learner with a problem-solving mindset, passionate about transforming complex challenges into practical, scalable solutions. Eager to contribute to challenging technical roles and grow within a dynamic organization.

Education

Vel Tech University

B.Tech, CSE (AI & Data Science), CGPA: 7.97

Aug 2023 – Jun 2027

Chennai, Tamilnadu

Experience

AICTE

May 2026 – Jul 2026

AI & Embedded Systems Intern

Chennai, Tamilnadu

- Hardware – Smart Waste Segregation Bin:** Built a 5-category waste classifier (wet, dry, plastic, biomedical, recyclable) via two-phase transfer learning on MobileNetV2 (frozen-head training, then fine-tuning the last 30 layers), achieving 97.76% validation accuracy.
- Merged and remapped two public Kaggle datasets (roughly 15K general-waste + biomedical/pharmaceutical images) into a unified 5-class scheme, using class-weighted loss and augmentation (brightness, rotation, shift) to handle class imbalance and flash-lit camera conditions.
- Deployed the trained model via a Flask inference server integrated with ESP32-CAM hardware, with a confidence-threshold fallback to a moisture sensor for low-confidence predictions.
- Software – Child Nutrition Tracking System:** Developed a cloud-based analytics platform (Next.js, Supabase, Vercel) for a government Child Development Department, applying predictive AI models to track malnutrition risk, health records, and welfare scheme delivery across child beneficiaries.

Projects

Telco Customer Churn MLOps Platform | Python, XGBoost, FastAPI, Docker, Kubernetes, MLflow

[GitHub](#)

- Built an end-to-end, reproducible MLOps system automating the journey from versioned customer data to monitored production inference and model promotion, achieving 0.847 ROC-AUC.
- Improved recall from 48% to 80% through threshold optimization; automated CI/CD (Jenkins), experiment tracking (MLflow/DVC), and drift-aware retraining with Prometheus/Grafana monitoring.

TN Government AI Scheme Assistant | FastAPI, Next.js, PostgreSQL, ChromaDB, RAG

[GitHub](#)

- Developed a bilingual (English/Tamil) RAG assistant consolidating 37+ government welfare scheme documents for citation-grounded benefit and eligibility lookup.
- Achieved a 100% pass rate on a 50-query benchmark and 86/86 validation tests; deployed live using Groq LLaMA 3.3 for real-time scheme discovery.

Autonomous Data Analyst (ADA) | Python, Multi-Agent System, AutoML, Streamlit, Flask

[GitHub](#)

- Built a 7-agent autonomous system automating missingness diagnosis, PII masking, EDA, and AutoML (LightGBM/XGBoost) tuning with 100% verified numerical grounding and zero arbitrary code execution.
- Automated executive PDF/PPTX report generation in under 25s, validated by a 590-test suite passing at 100%.

Technical Skills

Languages: Python, SQL, Java (Basics).

AI & Machine Learning: Deep Learning, Computer Vision, NLP, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Scikit-learn, TensorFlow, Keras, XGBoost, CNN.

Data Science & Analytics: Data Cleaning, Exploratory Data Analysis (EDA), Predictive Modeling and Analytics, Statistical Analysis, Model Evaluation.

MLOps & Deployment: Docker, Kubernetes, FastAPI, MLflow, CI/CD (Jenkins), Prometheus/Grafana.

Data Visualization: Power BI, Matplotlib, Seaborn, DAX.

Embedded AI & IoT: Arduino UNO, ESP32-CAM, Internet of Things (IoT), Sensor Integration, Embedded Systems.

Tools & Practices: GitHub, VS Code, Jupyter Notebook, Google Colab.

Certifications

RapidMiner (Altair) – Data Engineering Master Certification

Nov 2025

Deloitte – Data Analytics Job Simulation

Sep 2025

Tata Group – Data Visualisation: Empowering Business with Effective Insights

Sep 2025

Infosys Springboard – Exploring Data Visualization

Feb 2026